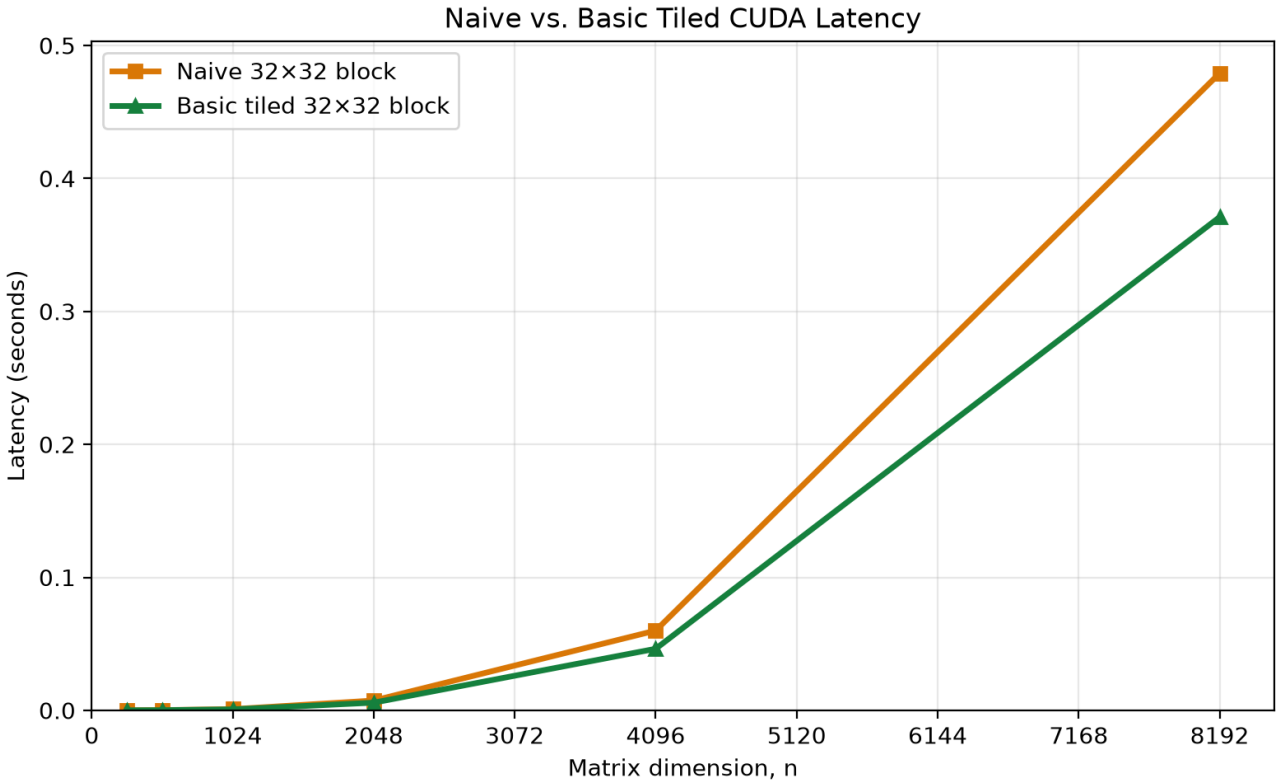
Section Overview

- Throughput, Arithmetic Intensity, and the Roofline Model

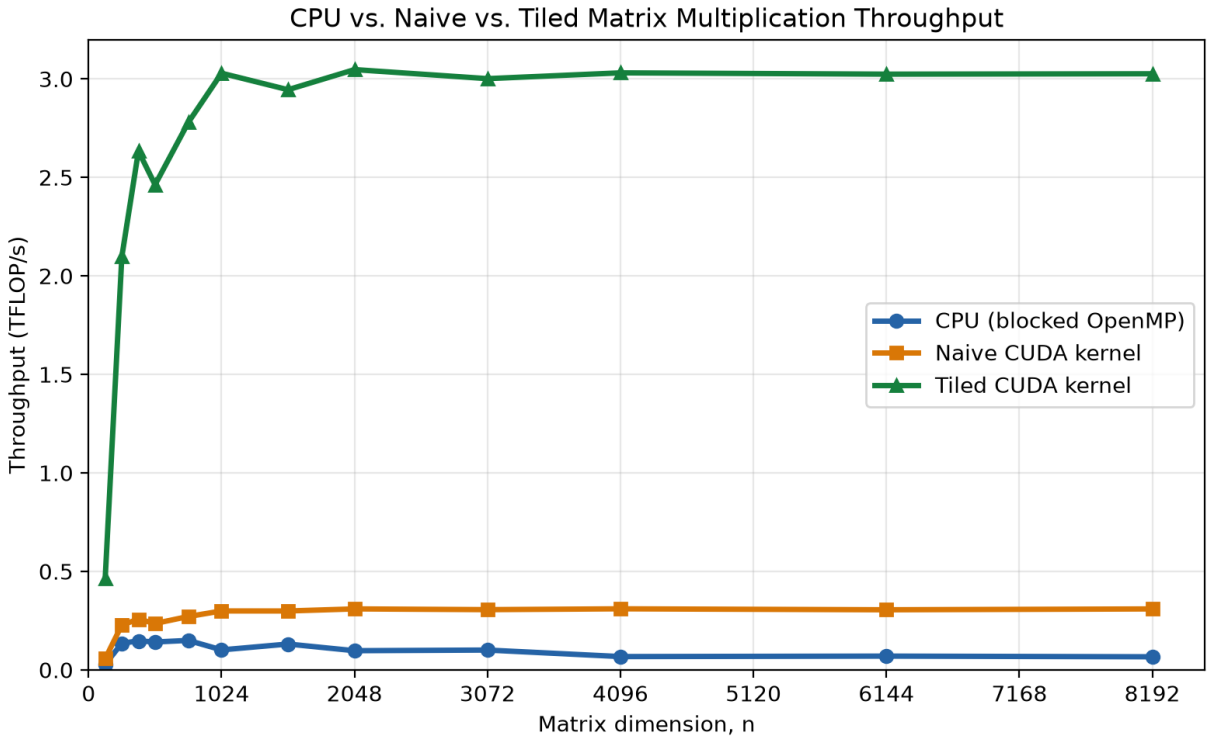
Why latency alone isn't enough

- Latency tells us how long one kernel launch takes, but it depends on the problem size.



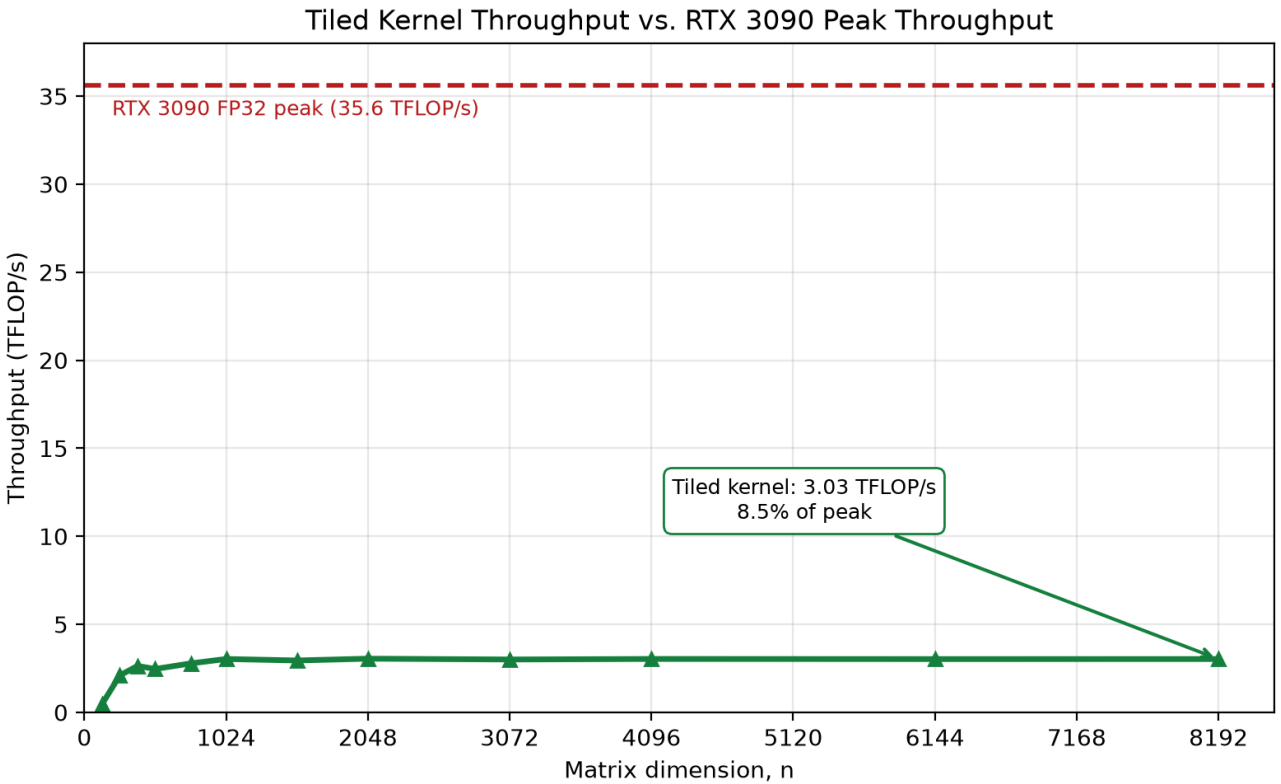
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Why throughput

Throughput tells us

1. A general performance metric, no matter the problem size
2. How well we are utilizing hardware

Throughput as a performance measure

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Example

Peak throughput of our RTX 3090 is 35.6 TFLOP/s. We observed 360ms or 0.36s of latency for $n = 8192$, where the total FLOPs required is 1.10 TFLOPs. Thus, we have an observed throughput of

$$\frac{1.10\text{TFLOPs}}{0.36\text{s}} \approx 3.05\text{TFLOP/s}.$$

Roughly 8.5% of peak throughput.

Arithmetic Intensity

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Example

For $n, k, m = 8192$, our tiled kernel moved a total of 137.7 GB of data from DRAM. With a total work count of 1.10 TFLOP, its arithmetic intensity is

$$\text{AI} = \frac{1.10 \times 10^{12} \text{ FLOP}}{137.7 \times 10^{9} \text{ bytes}} \approx \boxed{8.0 \text{ FLOP/byte}}.$$

Two Key Constraints of Hardware

Our hardware is characterized by

1. Peak throughput: 35.6 TFLOP/s
2. DRAM bandwidth: 935.8 GB/s

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Our tiling kernel has AI of 8.0 FLOP/byte, meaning it does 8 floating point operations for every byte of data moved from DRAM.
At 935.6 GB/s of bandwidth, our **memory roof** is

$$935.8 \text{ GB/s} \times 8.0 \text{ FLOP/byte} \approx 7.49 \text{ TFLOP/s}.$$

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This is much lower than the peak throughput of 35.6 TFLOP/s. **No matter how we optimize this kernel, without increasing AI, we will never achieve greater than 7.49 TFLOP/s**.

Hypothetical Kernel

Our hardware is characterized by

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Example

Imagine we have some kernel that has double the AI at 16.0 FLOP/s.

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$$935.8 \text{ GB/s} \times 16.0 \text{ FLOP/byte} \approx 15.0 \text{ TFLOP/s}.$$

Since $15.0 \text{ TFLOP/s} < 35.6 \text{ TFLOP/s}$, this kernel will still be memory bound.

Hypothetical Kernels

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1. Peak throughput: 35.6 TFLOP/s
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AI (FLOP/byte)	Upper bound throughput (TFLOP/s)
8	7.49
16	14.97
32	29.95
36	33.69
38	35.56
40	37.43

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38	35.56
40	37.43

Wait! Our peak throughput is 35.6 TFLOPs! At AI $\geq$ 38, we are **compute bound**. Our data can finally keep up with the CUDA Cores!

Ridge Point

Our hardware is characterized by

1. Peak throughput: 35.6 TFLOP/s
2. DRAM bandwidth: 935.8 GB/s

AI = 38 FLOP/byte is known as the **ridge point** *RP*, and can be found by solving:

$$\text{Data roof} = \text{Peak throughput}$$

$$\text{DRAM Bandwidth} \times \text{RP} = \text{Peak throughput}$$

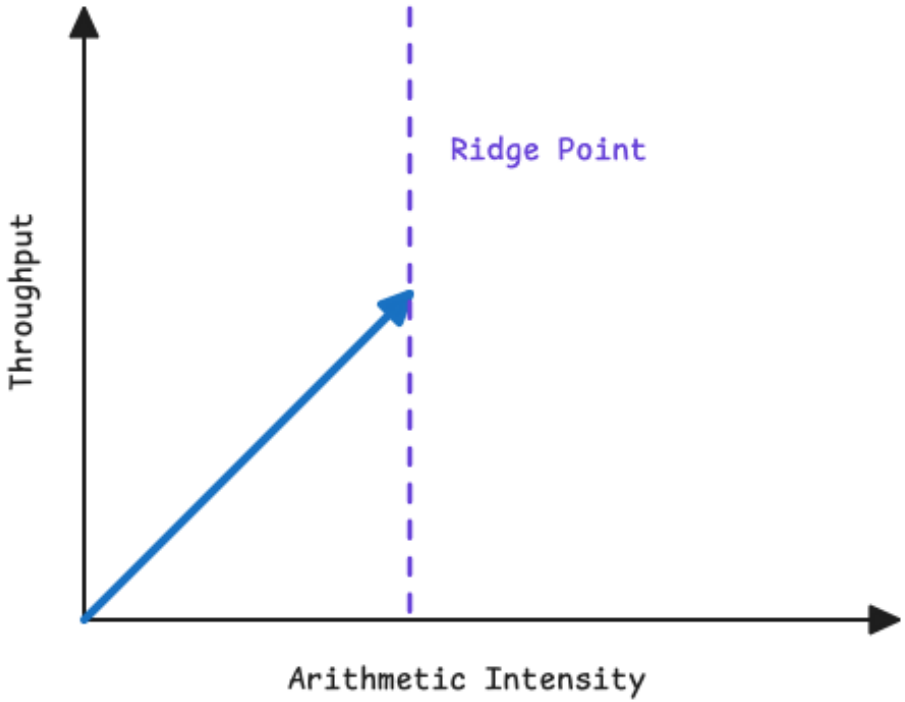
$$\text{RP} = \frac{\text{Peak throughput}}{\text{DRAM Bandwidth}}$$

$$\text{RP} = \frac{35.6 \text{ TFLOP/s}}{935.8 \text{ GB/s}} \approx 38.0 \text{ FLOP/s}$$

Roofline Model

Roofline Model

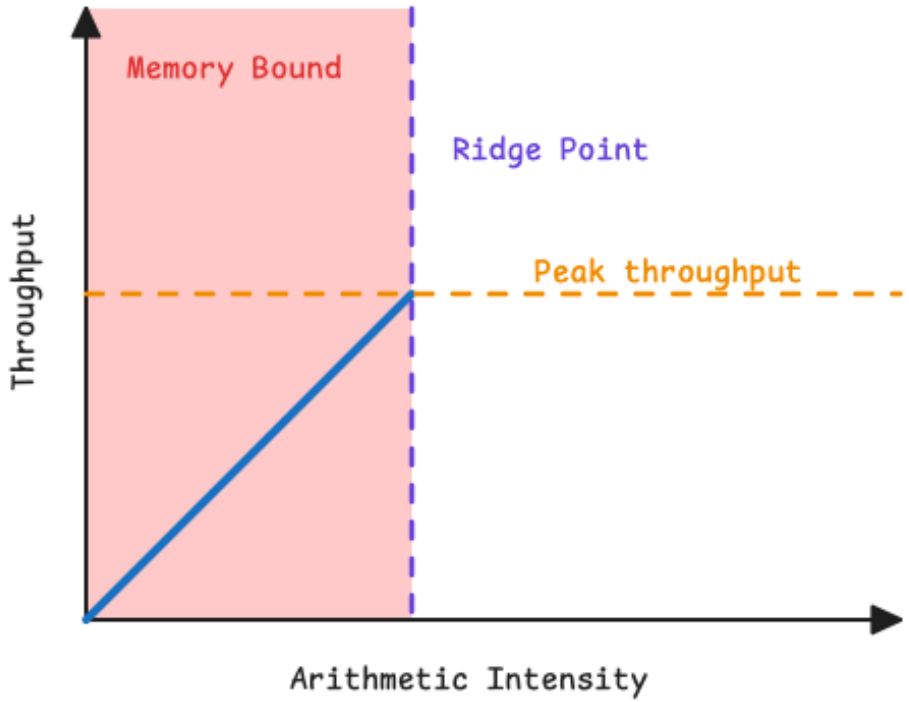
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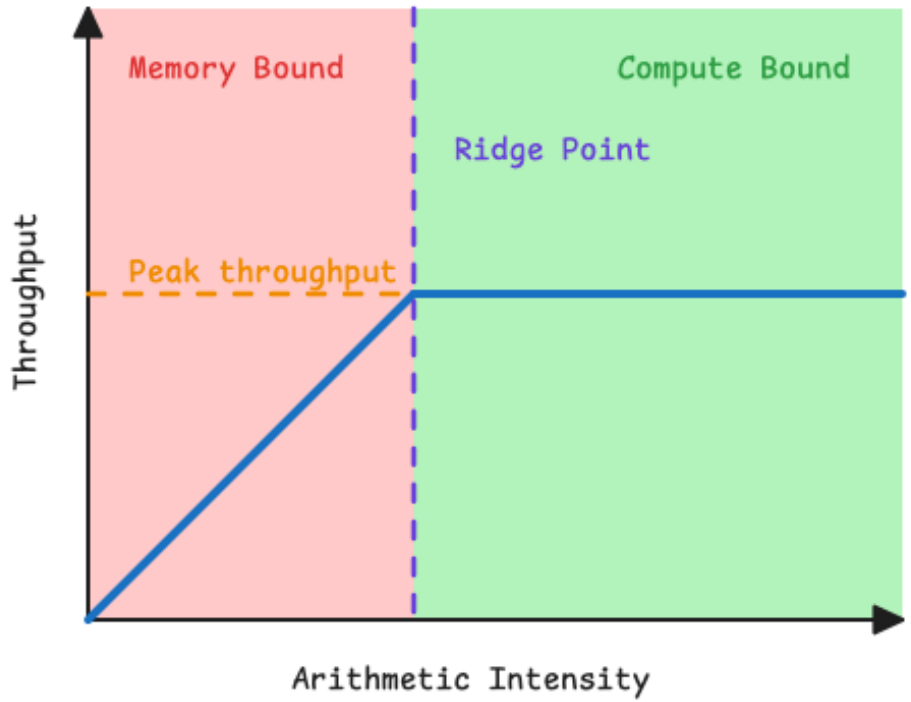
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Roofline Model

Roofline Model

For a given system, we can predict a the **upper bound** of throughput of a kernel given its AI. Up to the **ridge point**, we are **memory bound**, increasing AI will raise the upper bound of throughput. Past that, we are **compute bound**, increasing AI will not raise the upper bound of throughput.



Roofline Model of our RTX 3090

